Two vertex-disjoint monochromatic K_4s in every two-colouring of K_{20}

A candidate certificate-backed determination of $\mathrm{VR}_2(K_4)$

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Abstract

For a graph F without isolated vertices and a positive integer t, let $\mathrm{VR}_t(F)$ denote the least n such that every red–blue colouring of K_n contains t pairwise vertex-disjoint monochromatic copies of F; the copies need not have the same colour. We give a candidate computer-assisted proof that

$$\mathrm{VR}_2(K_4) = 20.$$

The lower bound is hand-checkable. It is supplied by a 19-vertex colouring obtained from the Paley graph P_{17} by replacing one vertex with three independent twins. For the upper bound, fix one homogeneous 4-set in a hypothetical counterexample on 20 vertices. The graph induced by the remaining 16 vertices must be a $(4,4)$-Ramsey graph, of which there are exactly two up to isomorphism. Each fixed core leaves 64 undetermined cross edges. Enumerating the 872 mixed four-sets that can be homogeneous and forbidding every disjoint pair gives a 64-variable CNF with 104,524 clauses. Both formulas have RUP/DRAT/LRAT-checked refutations. The paper separates the graph-theoretic reduction, the exact encoding equivalence, the proof-object checks, and the remaining trusted-computing assumptions.

Keywords: Ramsey theory; vertex-disjoint Ramsey number; monochromatic clique; exhaustive generation; SAT certificate; DRUP; LRAT.

2020 Mathematics Subject Classification: 05C55, 05C85.

Status. This is an **unrefereed candidate computer-assisted proof** offered for independent mathematical and computational scrutiny. The lower-bound construction is hand-verifiable. The two upper-bound proof objects have been replayed locally by separately implemented checkers, including a formally verified CakeML checking core. No external reproduction, human peer review, or end-to-end proof-assistant formalisation is claimed.

1 Introduction

For a graph F without isolated vertices and a positive integer t, the vertex-disjoint Ramsey number $\mathrm{VR}_t(F)$ is the least order of a complete graph whose every red–blue edge-colouring contains t pairwise vertex-disjoint monochromatic copies of F [2, 3]. Distinct copies may receive different colours. This distinguishes $\mathrm{VR}_t(F)$ from the ordinary symmetric Ramsey number of the disconnected graph tF, which requires a monochromatic union of all t copies.

Recent work has determined such numbers for triangles, paths, stars, matchings, and other graphs of small order or size [2, 3]. We consider K_4.

Theorem 1.1 (Main result)**.** *Every red–blue colouring of K_{20} contains two vertex-disjoint monochromatic copies of K_4, while some colouring of K_{19} does not. Equivalently,*

$$\boxed{\mathrm{VR}_2(K_4) = 20}.$$

The proof has two unequal parts. The lower bound is a direct construction based on the Paley graph P_{17}. The upper bound uses the classical identity $R(4,4) = 18$ and a complete catalogue of the two $(4,4)$-graphs on 16 vertices. Fixing either graph leaves a 64-variable propositional problem, and both resulting formulas are unsatisfiable.

The upper-bound computation originally appeared as the structural lemma inside a candidate proof concerning the cochromatic number of graphs on 20 vertices. The present paper extracts that lemma, supplies the sharp 19-vertex construction, and states the result in the established vertex-disjoint Ramsey notation. The cocolouring computations in the larger proof are not used here.

The logical architecture is as follows.

Component	Role
Lower construction	A hand-checkable colouring of K_{19} in which every monochromatic K_4 uses at least two vertices of a fixed three-vertex set.
Ramsey reduction	A counterexample on 20 vertices would consist of one homogeneous 4-set, one of the two 16-vertex $(4,4)$-cores, and 64 cross edges.
Exact encoding	The CNF is satisfied exactly by cross-edge assignments with no two disjoint homogeneous 4-sets.
Certificates	Separately implemented local checkers have accepted both fixed-core refutations.

2 Definitions and elementary invariance

Represent a red–blue colouring of K_n by a finite simple graph G: red edges are the edges of G, and blue edges are the edges of $\overline{G}$.

Definition 2.1. A vertex set $S \subseteq V(G)$ is *homogeneous* if $G[S]$ is complete or edgeless. Thus a homogeneous 4-set is exactly a monochromatic K_4 in the corresponding two-colouring. A $(4,4)$-graph is a graph containing neither a K_4 nor an independent set of order four.

Lemma 2.2 (Complement invariance)**.** *A vertex set is homogeneous in G if and only if it is homogeneous in $\overline{G}$. Consequently, the property of containing two vertex-disjoint homogeneous 4-sets is invariant under complementation.*

Proof. Complementation exchanges cliques and independent sets without changing their vertex sets. □

3 A sharp lower-bound colouring on nineteen vertices

Let

$$Q = \{1, 2, 4, 8, 9, 13, 15, 16\} \subset \mathbb{Z}_{17}$$

be the set of nonzero quadratic residues. The Paley graph P_{17} has vertex set $\mathbb{Z}_{17}$, with $i \sim j$ when $i - j \pmod{17} \in Q$ [6]. Since $-1 = 16 \in Q$, the graph is undirected.

Lemma 3.1. *The graph P_{17} contains neither a K_4 nor an independent set of order four.*

Proof. Translations and multiplications by nonzero squares are automorphisms of P_{17}, and they act transitively on its edges. A hypothetical K_4 may therefore be assumed to contain $\{0, 1\}$. The common neighbours of 0 and 1 are

$$Q \cap (1 + Q) = \{2, 9, 16\}.$$

Their pairwise differences are congruent, up to sign, to 7 or 14, neither of which lies in Q. They are therefore pairwise nonadjacent, so no two can complete $\{0, 1\}$ to a K_4.

Multiplication by a quadratic nonresidue, for example 3, is an isomorphism $P_{17} \cong \overline{P_{17}}$. The absence of a K_4 therefore also gives the absence of an independent four-set. □

We now replace vertex 9 by three independent twins. More precisely, start with $P_{17} - 9$, add a set

$$X = \{x_1, x_2, x_3\},$$

make X independent, and give every x_i the former external neighbourhood of vertex 9:

$$N = \{0, 1, 5, 7, 8, 10, 11, 13\}.$$

The remaining outside vertices are

$$M = \{2, 3, 4, 6, 12, 14, 15, 16\}.$$

Call the resulting graph $L_{19} = P_{17}[9 \rightsquigarrow \overline{K}_3]$.

Lemma 3.2. *The graph $P_{17}[N]$ is triangle-free, and $P_{17}[M]$ has no independent triple.*

Proof. The edges of $P_{17}[N]$ form the eight-cycle

$$0 - 8 - 7 - 5 - 1 - 10 - 11 - 13 - 0$$

together with the four opposite chords

$$01, \qquad 8\,10, \qquad 7\,11, \qquad 5\,13.$$

This is the eight-vertex Möbius ladder, hence is triangle-free; the claim can also be checked directly from the displayed edge list.

The edges of $\overline{P_{17}[M]}$ form the eight-cycle

$$2 - 12 - 15 - 4 - 16 - 6 - 3 - 14 - 2$$

together with

$$2\,16, \qquad 12\,6, \qquad 15\,3, \qquad 4\,14.$$

This graph is also triangle-free. Therefore $P_{17}[M]$ has no independent triple. □

Proposition 3.3. *The colouring represented by L_{19} contains no two vertex-disjoint monochromatic K_4s. Hence*

$$\mathrm{VR}_2(K_4) \geq 20.$$

Proof. There is no red K_4. A red K_4 wholly outside X would lie in $P_{17} - 9$, contradicting Lemma 3.1. A red K_4 using one vertex of X would require a triangle in N, contrary to Lemma 3.2. A red K_4 using at least two vertices of X is impossible because X is independent.

A blue K_4 wholly outside X would be an independent four-set in $P_{17} - 9$. A blue K_4 using exactly one vertex of X would require an independent triple in M. Both possibilities are excluded by Lemmas 3.1 and 3.2. Consequently every blue K_4 contains at least two of the three vertices of X. Any two such four-sets intersect. Thus the colouring contains no two vertex-disjoint monochromatic K_4s. □

Remark 3.4 (Sanity check). The graph L_{19} has 84 red edges and degree sequence $8^{11}10^8$. It contains no red K_4 and exactly 44 blue K_4s. Indeed, a blue copy uses either two twins and one of the 12 nonedges of $P_{17}[M]$, or all three twins and one vertex of M:

$$\binom{3}{2} \cdot 12 + \binom{3}{3} \cdot 8 = 44.$$

A direct verifier is included with the source bundle.

4 Reduction of the twenty-vertex upper bound

The classical Ramsey identity $R(4,4) = 18$ was established by Greenwood and Gleason [1]. It is also reproduced by the catalogue computation in section 6.

Lemma 4.1. *Every graph on* 20 *vertices contains two vertex-disjoint homogeneous* 4*-sets.*

Proof. Suppose that a graph G on 20 vertices contains no two disjoint homogeneous 4-sets. Since $20 \geq R(4,4) = 18$, the graph contains a homogeneous 4-set A. By Lemma 2.2, complementing G if necessary preserves the target property, so assume that A is a clique. Relabel

$$A = \{a_0, a_1, a_2, a_3\}, \qquad B = V(G) \setminus A.$$

If $G[B]$ contains a homogeneous 4-set, it is disjoint from A and the conclusion follows. Thus a counterexample must have $G[B]$ a $(4,4)$-graph on 16 vertices.

The complete catalogue in section 6 contains exactly two such graphs up to isomorphism. In graph6 notation they are

$$C_0 = \texttt{OsHHirKdlp[IFVI|KpqfR}, \qquad C_1 = \texttt{Ov?IXZIhlRWjUXL[iphst}.$$

Each has 60 edges, degree sequence $7^8 8^8$, 56 triangles, and 56 independent triples.

Once $G[B]$ is fixed to C_0 or C_1, the only undetermined edges are the $4 \cdot 16 = 64$ edges between A and B. For each core, section 5 constructs a CNF whose satisfying assignments are exactly those cross-edge patterns for which the resulting graph contains no two disjoint homogeneous 4-sets. Both formulas are unsatisfiable by the certificate checks in section 7. This contradiction proves the lemma. □

5 The exact fixed-core encoding

Fix $A = \{a_0, a_1, a_2, a_3\}$ as a clique and let $B = \{b_0, \ldots, b_{15}\}$ induce one of the two cores. For $0 \leq i < 4$ and $0 \leq j < 16$, introduce a Boolean variable

$$x_{i,j} = 1 \quad \Longleftrightarrow \quad a_i b_j \in E(G).$$

Lemma 5.1 (Mixed-set classification)**.** *The mixed four-sets that can be homogeneous are exactly the four types in table 1.*

Table 1: Conditions under which a mixed four-set is homogeneous.

Split	Necessary and sufficient condition	Count
$3A + 1B$	Clique: the three relevant cross edges are present.	$\binom{4}{3} \cdot 16 = 64$
$2A + 2B$	Clique: the B-pair is one of the 60 core edges and all four relevant cross edges are present.	$\binom{4}{2} \cdot 60 = 360$
$1A + 3B$	Clique: the B-triple is one of the 56 triangles and all three cross edges are present.	$4 \cdot 56 = 224$
$1A + 3B$	Independent: the B-triple is one of the 56 independent triples and all three cross edges are absent.	$4 \cdot 56 = 224$
	Total	**872**

Proof. A four-set containing three vertices of A already contains a triangle, so it can only be a clique; this occurs exactly when its three cross edges are present. A four-set containing two vertices of A contains their edge and therefore cannot be independent. It is a clique exactly when the B-pair is an edge and all four cross edges are present.

With one vertex in A, the B-triple must be either a triangle, followed by three present cross edges, or an independent triple, followed by three absent cross edges. No four-set wholly in B is homogeneous, and A is the only four-set wholly in A. □

Represent each possible mixed homogeneous set S by the conjunction C_S of its three or four required literals. Whenever $S \cap T = \varnothing$, add the clause

$$\neg C_S \vee \neg C_T.$$

For example, if

$$S = \{a_0, a_1, a_2, b_5\}, \qquad T = \{a_3, b_1, b_4, b_9\},$$

and $\{b_1, b_4, b_9\}$ is an independent triple of the core, the clause is

$$\neg x_{0,5} \vee \neg x_{1,5} \vee \neg x_{2,5} \vee x_{3,1} \vee x_{3,4} \vee x_{3,9}.$$

Proposition 5.2 (Encoding equivalence)**.** *For either core, an assignment to the* 64 *variables satisfies the constructed CNF if and only if the corresponding* 20*-vertex graph has no two vertex-disjoint homogeneous* 4*-sets.*

Proof. By Lemma 5.1, every mixed set that can be homogeneous is represented by exactly one conjunction C_S, and C_S holds exactly when S is homogeneous. The clause belonging to disjoint sets S, T is false exactly when both sets are homogeneous. Conjoining the clauses over all disjoint pairs therefore forbids exactly, and only, the existence of two disjoint mixed homogeneous four-sets.

In the branch being encoded, B contains no homogeneous four-set, so A cannot be paired with a disjoint homogeneous set. Every prohibited pair is therefore covered by the mixed-set clauses. □

For each core, the resulting DIMACS formula has 64 variables and 104,524 clauses. Its clause-length histogram is

43,840 of length 6, 52,416 of length 7, 8,268 of length 8.

6 Catalogue generation

McKay's maintained Ramsey catalogue lists two non-isomorphic $(4,4)$-graphs of order 16 [4]. The proof bundle also regenerates the catalogue from the definition, using nauty for canonical labelling [5].

6.1 Completeness of the generation scheme

Suppose a complete set of isomorphism-class representatives is known at order n. Every $(4,4)$-graph H on $n+1$ vertices remains a $(4,4)$-graph after deleting a vertex. Therefore H arises from an order-n representative by adjoining one vertex with some neighbourhood S. Trying every admissible S, canonicalising each extension, and retaining one copy of every canonical form reaches every order-$n+1$ class. Induction begins with the unique one-vertex graph.

For a new vertex with neighbourhood S, a new homogeneous four-set must contain that vertex. The extension is admissible exactly when

$$G[S] \text{ is triangle-free} \quad \text{and} \quad G[V \setminus S] \text{ has no independent triple.}$$

The implementation restricts to $n - 8 \le |S| \le 8$. This pruning follows from the elementary bound that a triangle-free graph with independence number at most three has at most eight vertices. One proof uses $R(3,3) \le 6$: in a hypothetical nine-vertex example every degree is at most three; a vertex of degree at most two would leave six non-neighbours, which contain an independent triple, while all degrees equal to three is impossible by the handshaking lemma. Applying the same argument to the complement gives the lower bound on $|S|$.

6.2 Replayed output

The complete run was replayed on 26 July 2026. The numbers of non-isomorphic $(4,4)$-graphs were

n	1	2	3	4	5	6	7	8	9
#	1	2	4	9	24	84	362	2,079	14,701

n	10	11	12	13	14	15	16	17	18
#	103,706	546,356	1,449,166	1,184,231	130,816	640	2	1	0.

The two order-16 outputs agree character-for-character with the catalogue strings C_0, C_1. The single survivor at order 17 and empty level at order 18 locally reproduce $R(4,4) = 18$.

7 Certificate verification

For each fixed core, a clause-learning-free DPLL program produced a tree-like DRUP refutation. Core 0 visited 1,571 search nodes, made 785 decisions, and emitted 1,571 proof lines.

Core 1 visited 789 nodes, made 394 decisions, and emitted 789 proof lines. The preserved formulas have SHA-256 hashes

C_0 : `cbfea7b0b2cb712ad8b4f8b129f74c7f010e879064414eaef76e74aa88be3f44`,

C_1 : `d3d96a3c2a719c753242673b81aabc3977c5e435bb5996b4c58c85a82d11dc5a`.

The following checks were replayed against formulas regenerated separately from the two graph6 strings.

Table 2: Replayed validation of the fixed-core refutations.

Check	Core 0	Core 1
Separate formula construction	872 conditions; 104,524 clauses; clause set identical	872 conditions; 104,524 clauses; clause set identical
Direct semantic spot-check	300/300 assignments agreed	300/300 assignments agreed
Separate RUP checker	1,571 steps; empty clause derived	789 steps; empty clause derived
Package RUP checker	`VERIFIED`	`VERIFIED`
`drat-trim` [7]	`s VERIFIED`; 0 RAT lemmas	`s VERIFIED`; 0 RAT lemmas
`cake_lpr` [8]	`s VERIFIED UNSAT`	`s VERIFIED UNSAT`

The separately regenerated formulas were byte-identical to the packaged formulas. The package audit also tested 1,000 deterministic assignments per core, and the twelve-entry SHA-256 manifest passed in full. The DRUP proofs were converted to LRAT with `drat-trim -L`; the resulting LRAT proofs were accepted against the original CNFs by `cake_lpr`, whose checking core is formally verified in CakeML [8].

Cross-pairing the first formula with the second proof was rejected by the standalone RUP checker, by `drat-trim`, and by `cake_lpr`. One operational detail matters: `cake_lpr` can print a checking failure while returning process status zero, so automated replay must require the exact verdict `s VERIFIED UNSAT`.

8 Completion of the proof and a higher-copy consequence

Proof of theorem 1.1. The construction in Proposition 3.3 gives $\mathrm{VR}_2(K_4) \geq 20$. By Lemma 4.1, every colouring of K_{20} has two vertex-disjoint monochromatic K_4s. Hence $\mathrm{VR}_2(K_4) \leq 20$. □

$$\boxed{\mathrm{VR}_2(K_4) = 20}$$

The exact two-copy value immediately improves the standard greedy bound for every larger number of copies.

Corollary 8.1. *For every integer $t \geq 2$,*

$$\mathrm{VR}_t(K_4) \leq 4t + 12.$$

Proof. The case $t = 2$ is Theorem 1.1. For $t \geq 3$, consider a colouring of K_{4t+12}. Since $4t + 12 \geq R(4, 4) = 18$, choose and remove one monochromatic K_4. The remaining graph has order $4(t - 1) + 12$, so induction supplies $t - 1$ further pairwise disjoint monochromatic copies. This is the general extension principle recorded in [2]. □

In particular, $\mathrm{VR}_3(K_4) \leq 24$. A separate classification of the 19-vertex two-copy extremals sharpens the three-copy problem further, but that classification and its additional certificates are not needed for the present theorem.

9 Assurance boundary and reproducibility

The certificate checks establish unsatisfiability of two exact CNFs. The full graph-theoretic theorem also uses the catalogue and the semantic bridge from graphs to those formulas. The layers should therefore be distinguished.

Layer	What is established, and what remains trusted
Hand mathematics	The Paley construction, lower-bound proof, complement invariance, catalogue-completeness induction, mixed-set classification, encoding equivalence, and final deductions can be checked directly from the text.
SAT proof objects	The two DRUP refutations are accepted by separately implemented local RUP checkers and by `drat-trim`; their LRAT translations are accepted against the exact CNFs by the formally verified `cake_lpr` checking core.
Semantic bridge	Proposition 5.2 proves the intended equivalence, and a separate direct graph implementation agrees on deterministic test assignments. The source implementation is not formally verified.
Catalogue	The generation scheme is exhaustive if its implementation and nauty canonicalisation are correct. Its full sequence matches McKay's maintained data, but the C generator and nauty are not proved correct in a proof assistant.
System	The compiler, operating system, file I/O, and hardware remain in the conventional trusted computing base, except where the CakeML checker reduces that base for the SAT-checking layer.

The replay record and this companion are bound to the following artifacts.

- Current archival dependency: https://github.com/ipitchford/z20-cochromatic, annotated tag `candidate-2026-07-26`, commit `3c7e520fdc0615f5c700761c2b1e5108dcc836e7`.
- Historical replay provenance: local source commit `61e8300`. The dated receipt records that the replay originally reviewed public snapshot `8d9829edebecdfc78384eef5b41617ddd32b11ea`; the load-bearing source and proof objects are byte-identical at the archival dependency above.
- Replay date and platform: 26 July 2026, macOS on arm64.
- Replay receipt: `paper/z20_equals_6_replay_receipt_2026-07-26.md`.

At the pinned commit, the load-bearing public paths are:

- catalogue: `src/gen44.c` and `src/gen44_result.txt`;
- formula generator: `certificates/generate.py`;
- core 0: `certificates/z20_core0.cnf`, `certificates/z20_core0.drup`, and `certificates/core0.lrat`;
- core 1: `certificates/z20_core1.cnf`, `certificates/z20_core1.drup`, and `certificates/core1.lrat`;
- package checks: `certificates/audit.py` and `certificates/check_drup.c`; and
- separate semantic/RUP replay: `verification/verify_certificate.py`.

The companion directory contains an exact path-and-hash map and a small direct verifier for the 19-vertex lower construction. Nauty, DRAT-trim, and `cake_lpr` are not vendored.

Status. The accurate present claim is: *a complete, locally regenerated candidate computer-assisted determination of* $\mathrm{VR}_2(K_4) = 20$, *with a hand-checkable lower bound, certificate-backed fixed-core refutations, and an explicit trusted- computing boundary*.

10 Provenance and AI disclosure

The source repository reports that the underlying mathematical and computational work was produced by AI systems, while the human participant selected the problem, directed the work, preserved the artifacts, and mediated between systems. Ian Pitchford is the repository maintainer and publisher and performed that problem-selection and research-mediation role. The reported division of labour is: an earlier Hermes-driven search; subsequent instrumentation and survivor analysis; the two-core reduction, CNF generator, DPLL solver, and DRUP certificates from GPT-5.6 Sol; and separate verification, catalogue regeneration, and earlier exposition by Claude Opus 5. The present companion paper was prepared for repository publication with OpenAI Codex from the preserved proof manuscript, replay receipt, source code, certificates, and the sharp lower-bound construction. This paragraph records project provenance; it is not an externally audited authorship determination.

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A Exact lower-bound witness data

With vertices ordered as

$$0, 1, \ldots, 8, 9, 10, \ldots, 16, 17, 18,$$

where $9, 17, 18$ are the three independent twins, one graph6 encoding of L_{19} is

```
RzlXWmJpZDeJEJbDgp\EJsWnDlEJY ?.
```

The backslash is a literal graph6 character. The direct verifier supplied with the paper reconstructs the graph from the Paley rule rather than trusting this string. It enumerates all $\binom{19}{4} = 3{,}876$ four-sets and confirms:

red edges	84,
red K_4s	0,
blue K_4s	44,
vertex-disjoint monochromatic pair	none.

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